

Topic

- Hyperparameters vs Parameters
 - Why tuning is required
 - KNN & Decision Tree tuning
 - GridSearchCV (main focus)
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Class Objective

Students will:

- Understand what **hyperparameters** are
 - Learn why default model settings are not optimal
 - Tune models using **GridSearchCV**
 - Improve model performance practically
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1. CLASS START (Hook – VERY IMPORTANT)

Start like this:

“Kal humne KNN model banaya...

But ek important question — K = 3 hi kyu?

K = 5 ya 7 kyu nahi?”

 Students think → curiosity build → perfect entry

2. WHAT ARE PARAMETERS VS HYPERPARAMETERS

◆ Parameters

- Learned by model automatically
 - Example:
 - Linear Regression → coefficients
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◆ Hyperparameters

- Set by user before training
 - Control model behavior
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Examples

Algorithm	Hyperparameter
KNN	n_neighbors (K)

Algorithm	Hyperparameter
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Decision Tree	max_depth
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Logistic Regression	C
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💡 **Teaching Line**

“Hyperparameters are the settings we choose before training.”

3. 🚫 **WHY HYPERPARAMETER TUNING?**

Problem

Default values:

- Not always best
 - May cause:
 - Overfitting
 - Underfitting
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Goal

👉 Find best combination of parameters

💡 **Teaching Line**

“Better parameters = better model performance”

4. 📊 **EXAMPLE (VERY IMPORTANT)**

KNN

- $K = 1 \rightarrow$ overfit
- $K = 10 \rightarrow$ underfit

👉 Best K lies somewhere in between

Decision Tree

- Small depth $\rightarrow$ underfit
 - Large depth $\rightarrow$ overfit
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5. 🇮🇹 TYPES OF TUNING (Brief)

◆ Manual Tuning

- Try values manually
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◆ Grid Search (Main Focus ✅)

- Try all combinations
 - Select best
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◆ Random Search (just mention)

6. 📖 WHAT IS GRIDSEARCHCV?

Definition

GridSearchCV automatically tries multiple hyperparameter combinations and selects the best one using cross-validation.

Simple Meaning

“It tests many parameter values and gives the best one.”

7. 🔄 HOW IT WORKS

1. Define parameter grid
 2. Try all combinations
 3. Apply cross-validation
 4. Select best result
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💡 Teaching Line

“GridSearchCV = tuning + cross validation together”

8. 🎯 INTERVIEW CONCEPTS

Q1: What is hyperparameter?

👉 Parameter set before training.

Q2: Why tuning needed?

👉 Default values may not be optimal.

Q3: What is GridSearchCV?

👉 Tool to find best hyperparameters.

Q4: Difference between parameter & hyperparameter?

👉 Parameter learned, hyperparameter set manually.

OUTPUT INTERPRETATION

Explain:

- `best_params_` → best K value
 - `best_score_` → accuracy
 - Model improved automatically
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👉 Example:

“Best K = 5 gives highest accuracy”

A model is not powerful by default — it becomes powerful when we tune it correctly